

# Exam 2 Review: Choosing the Right Geometric Tool

Tuesday, November 17, 2026

## Visualization: State space determines the first calculation

| state space                    | first calculation                                             | global structure                                                             |
|--------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------|
| $S^1$                          | $f(\theta) = 0$ and $\text{sign } f$                          | $f \neq 0$ :<br>$T = \left  \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $ |
| $\mathbb{R}^2, \dot{z} = Az$   | $\tau = \text{tr } A, \Delta = \det A,$<br>$\tau^2 - 4\Delta$ | eigendirections and orientation                                              |
| $\mathbb{R}^2, \dot{z} = F(z)$ | $F(z) = 0, DF(z_*)$                                           | nullclines, invariant sets,<br>energy, symmetry                              |
| $S^1 \times \mathbb{R}$        | $F(z) = 0, DF(z_*)$                                           | periodic identification and<br>energy levels, when available                 |

## Observation: Scope of nonlinear linearization

At a hyperbolic equilibrium of a  $C^1$  nonlinear planar system, the nonlinear flow is locally topologically equivalent to its linearization. In particular, the linearization correctly determines whether the equilibrium is a sink, a source, or a saddle. At a nonhyperbolic equilibrium, the theorem is inconclusive.

## Question: A phase-locking threshold on the circle

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1,$$

find and classify all fixed points for every  $\mu \in \mathbb{R}$ .

**Solution.** Let

$$f(\theta) = \mu - \sin \theta.$$

If  $|\mu| < 1$ , define

$$\alpha = \arcsin \mu, \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The two fixed points modulo  $2\pi$  are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -\cos \theta,$$

**Question: A phase-locking threshold on the circle (continued)**

we have

$$f'(\theta_s) < 0, \quad f'(\theta_u) > 0.$$

Thus  $\theta_s$  is asymptotically stable and  $\theta_u$  is unstable. If  $|\mu| > 1$ , there are no fixed points, and every solution rotates around the circle. At  $\mu = 1$  the fixed points coalesce at  $\theta = \pi/2$ ; at  $\mu = -1$  they coalesce at  $\theta = 3\pi/2$ . In both cases

$$f = 0, \quad f_\theta = 0, \quad f_{\theta\theta} \neq 0, \quad f_\mu = 1 \neq 0,$$

so each threshold is a saddle-node bifurcation on  $S^1$ . At  $\mu = 1$  the fixed point attracts from the left and repels from the right; at  $\mu = -1$  it repels from the left and attracts from the right. Therefore each threshold fixed point is half-stable and Lyapunov unstable.

**Question: Rotation period beyond phase locking**

For

$$\dot{\theta} = \mu - \sin \theta, \quad \theta \in S^1, \quad |\mu| > 1,$$

compute the positive rotation period.

**Solution.** If  $\mu > 1$ , then  $\mu - \sin \theta > 0$  and

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\mu - \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\mu - \cos \phi} \quad (\phi = \theta - \pi/2) \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\mu - \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 ds}{(\mu - 1) + (\mu + 1)s^2} \quad \left(s = \tan \frac{\phi}{2}\right) \\ &= \frac{2\pi}{\sqrt{\mu^2 - 1}}. \end{aligned}$$

If  $\mu < -1$ , the orientation reverses and the positive period is again

$$T = \frac{2\pi}{\sqrt{\mu^2 - 1}}.$$

**Notation: Trace-determinant data**

For a real  $2 \times 2$  matrix  $A$ , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

Here  $D$  is the discriminant of the characteristic polynomial

$$\lambda^2 - \tau\lambda + \Delta.$$

**Visualization: Trace–determinant classification**For  $\dot{z} = Az$ ,

| condition                     | type                         |
|-------------------------------|------------------------------|
| $\Delta < 0$                  | saddle                       |
| $\Delta > 0, D > 0, \tau < 0$ | asymptotically stable node   |
| $\Delta > 0, D > 0, \tau > 0$ | unstable node                |
| $\Delta > 0, D < 0, \tau < 0$ | asymptotically stable spiral |
| $\Delta > 0, D < 0, \tau > 0$ | unstable spiral              |
| $\Delta > 0, D < 0, \tau = 0$ | linear center                |

**Recall: Degenerate linear cases**

When  $D = 0$  and  $\Delta > 0$ , eigenvectors distinguish a star from a degenerate node. When  $\Delta = 0$ , the kernel of  $A$  is nontrivial, so the linear system has a non-isolated set of equilibria.

**Question: Recover a complete linear portrait**

Classify and orient

$$\dot{z} = Az, \quad A = \begin{pmatrix} 1 & -2 \\ 2 & 1 \end{pmatrix}.$$

**Solution.** We compute

$$\tau = 2, \quad \Delta = 5, \quad D = 2^2 - 4(5) = -16.$$

The eigenvalues are

$$\lambda = 1 \pm 2i,$$

so the origin is an unstable spiral. At  $(1, 0)$ ,

$$A \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix},$$

which points upward; the rotation is counterclockwise. Since

$$A = I + 2 \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

the solution operator is

$$e^{At} = e^t \begin{pmatrix} \cos 2t & -\sin 2t \\ \sin 2t & \cos 2t \end{pmatrix}.$$

Thus radii grow like  $e^t$  and the angular velocity is 2.

**Lemma: An invariant zero set is a uniqueness barrier**

Let  $G, H : \mathbb{R}^2 \rightarrow \mathbb{R}$  be locally Lipschitz, and consider the system

$$\dot{x} = xG(x, y), \quad \dot{y} = H(x, y).$$

Then the line  $x = 0$  is invariant. No trajectory can cross from  $x > 0$  to  $x < 0$ .

**Proof.** Let  $y(t)$  solve

$$\dot{y} = H(0, y), \quad y(t_0) = y_0.$$

Then  $(0, y(t))$  solves the full system through  $(0, y_0)$ . Local uniqueness forces every solution through that state to remain on the axis. A trajectory crossing from  $x > 0$  to  $x < 0$  would meet the axis and then leave it, which is impossible.  $\square$

**Question: When does an imaginary linear pair give a nonlinear center?**

Analyze

$$\dot{x} = y, \quad \dot{y} = -x - x^3.$$

Determine the global phase curves and classify the origin.

**Solution.** The origin is the only equilibrium. Its Jacobian is

$$DF(0, 0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

with eigenvalues  $\pm i$ , so linearization alone is inconclusive. Define

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4.$$

Then

$$\begin{aligned} \dot{E} &= y\dot{y} + (x + x^3)\dot{x} \\ &= y(-x - x^3) + (x + x^3)y \\ &= 0. \end{aligned}$$

For every  $h > 0$ , there is a unique number  $A_h > 0$  satisfying

$$\frac{1}{2}A_h^2 + \frac{1}{4}A_h^4 = h.$$

The level  $E = h$  consists of the two graph branches

$$y = \pm \sqrt{2h - x^2 - \frac{1}{2}x^4}, \quad |x| \leq A_h,$$

which meet only at  $(\pm A_h, 0)$ . Hence the level is a compact simple closed curve. Moreover,

$$\nabla E = (x + x^3, y)$$

vanishes only at the origin, which does not lie on a positive level. Therefore every level  $E = h$ ,  $h > 0$ , is a periodic orbit, and the origin is a nonlinear center.

**Question: Period of the nonlinear oscillator**

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3,$$

set up the period of the orbit through  $(A, 0)$ , where  $A > 0$ .

**Solution.** For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

the orbit through  $(A, 0)$  has energy

$$E_A = \frac{1}{2}A^2 + \frac{1}{4}A^4.$$

On the branch with  $y > 0$ ,

$$y = \sqrt{2E_A - x^2 - \frac{1}{2}x^4}, \quad dt = \frac{dx}{y}.$$

Reflection symmetry about both coordinate axes makes the four quarter-orbit travel times equal. Hence

$$T(A) = 4 \int_0^A \frac{dx}{\sqrt{2E_A - x^2 - \frac{1}{2}x^4}}.$$

With  $x = A \sin \phi$ ,

$$2E_A - x^2 - \frac{1}{2}x^4 = A^2 \cos^2 \phi \left( 1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi \right),$$

so

$$T(A) = 4 \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 + \frac{A^2}{2} + \frac{A^2}{2} \sin^2 \phi}}.$$

In particular,

$$\lim_{A \rightarrow 0} T(A) = 2\pi.$$

**Question: What changes when linear damping is added?**

For

$$\dot{x} = y, \quad \dot{y} = -x - x^3 - \varepsilon y, \quad \varepsilon > 0.$$

Compute  $\dot{E}$  and determine whether the damped system can have a nonconstant periodic orbit.

**Solution.** For

$$E(x, y) = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4,$$

we obtain

$$\dot{E} = y(-x - x^3 - \varepsilon y) + (x + x^3)y = -\varepsilon y^2 \leq 0.$$

If a nonconstant solution were periodic with period  $T$ , then

$$0 = E(x(T), y(T)) - E(x(0), y(0)) = -\varepsilon \int_0^T y(t)^2 dt,$$

**Question: What changes when linear damping is added? (continued)**

so  $y \equiv 0$ . The equation

$$\dot{y} = -x - x^3 = -x(1 + x^2)$$

then forces  $x \equiv 0$ . Therefore the damped system has no nonconstant periodic orbit.

**Recall: Regular level set**

Let  $U \subseteq \mathbb{R}^2$  be open, let  $E \in C^1(U)$ , and let  $c \in \mathbb{R}$ . The level set  $E = c$  is **regular** if

$$\nabla E(z) \neq 0 \quad \text{at every } z \in U \text{ satisfying } E(z) = c.$$

**Visualization: Hypotheses and their valid conclusions**

Let  $U \subseteq \mathbb{R}^2$  be open and let  $F : U \rightarrow \mathbb{R}^2$  be a vector field.

| calculation                                                                                                                                                                | valid conclusion                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| $z_*$ is an equilibrium,<br>$DF(z_*)$ has eigenvalues $\pm i$<br>$H \in C^1(U)$ , $H^{-1}(0)$ regular,<br>$\nabla H \cdot F = 0$ on $H^{-1}(0)$ ,<br>$F$ locally Lipschitz | linearization is inconclusive                                       |
| $E \in C^1(U)$ , $\dot{E} = 0$                                                                                                                                             | $H^{-1}(0)$ is invariant                                            |
| $E \in C^1(U)$ ,<br>$\dot{E} < 0$ away from equilibria                                                                                                                     | each trajectory stays in one connected component of an energy level |
| $F$ locally Lipschitz                                                                                                                                                      | no nonconstant periodic orbit                                       |
|                                                                                                                                                                            | solution curves that meet agree after a time shift                  |

**Visualization: Invalid converses and overclaims**

Let  $F$  be the vector field, and let  $H$  and  $E$  be differentiable real-valued functions on its phase space.

$$\lambda = \pm i \not\Rightarrow \text{nonlinear center,}$$

$$H^{-1}(0) \text{ invariant} \not\Rightarrow H^{-1}(0) \text{ is the global basin boundary,}$$

$$\dot{E} = 0 \not\Rightarrow \text{every connected component of an energy level is a periodic orbit,}$$

$$\text{numerical portrait} \not\Rightarrow \text{proof.}$$